DP Biology – C1.1

Enzymes and metabolism

- How do enzymes act on, and respond to, other molecules?
- Which parts of metabolism depend on one another?

SL statements C1.1.1–10. Textbook pp. 380–395.

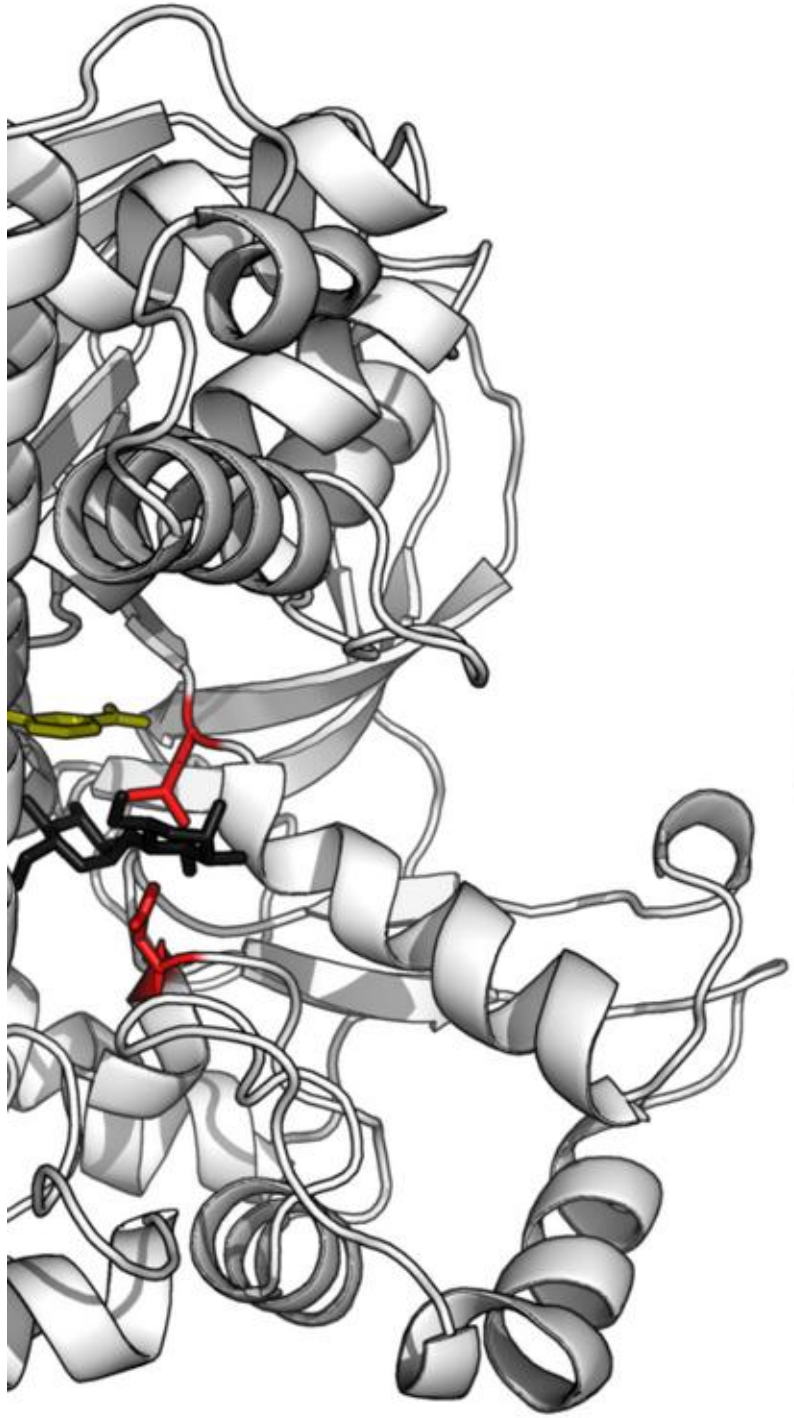


Image: model of a glucosidase enzyme. Thomas Shafee, CC BY 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	C1.1.1–3	Explain why cells need enzymes, and distinguish anabolic and catabolic reactions in metabolism
2	C1.1.4–7	Explain how the active site, induced fit and collisions lead to catalysis, and why enzymes are specific
3	C1.1.8–9	Explain how temperature, pH and the amount of substrate change the rate, and calculate rates from data
4	C1.1.10	Explain how enzymes lower activation energy, and apply the ideas to an experiment

The big picture: how an enzyme works



Every cell runs thousands of reactions at once. Almost every one is catalysed by its own **enzyme**, a protein whose **active site** fits its substrate.

Because enzymes are proteins, their shape, and so their activity, depends on **temperature** and **pH**. Their rate also depends on how often they meet their substrate. This chapter explains each of these ideas in turn.

Big question for this chapter

Why does a fever of 42 °C put a person's life at risk, when 37 °C is fine?

Enzymes are catalysts

C1.1.1

A **catalyst** speeds up a chemical reaction without being used up, so it can be used again and again. **Enzymes** are biological catalysts.

Without enzymes, most reactions in cells would happen far too slowly at body temperature to keep a cell alive. Heating the cell would speed them up, but would also damage it.

Enzymes make reactions fast enough at the temperature and pH of the cell, and let the cell choose which reactions happen, by making only the enzymes it needs.

How much faster?

One enzyme in nucleotide synthesis (OMP decarboxylase) speeds its reaction from a half-life of about 78 million years to a few milliseconds. Even ordinary enzymes speed reactions up millions of times.

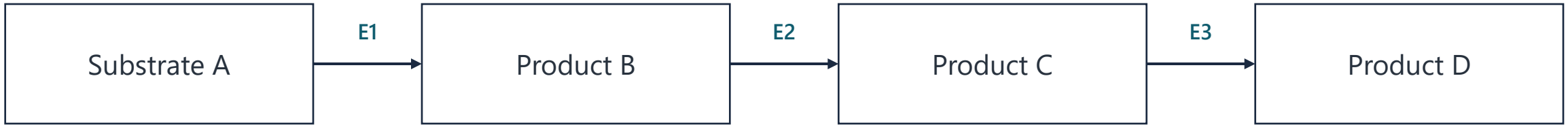
Key idea

Enzymes do not make impossible reactions happen; they make possible reactions happen fast enough for life.

Metabolism: a web of linked reactions

C1.1.2

Metabolism is the whole network of chemical reactions in a cell or organism. Many reactions are linked in **pathways**, where the product of one reaction is the substrate of the next, and each step is catalysed by its own **specific** enzyme.



Why it matters

Because each step needs its own enzyme, a cell can switch a whole pathway on or off by controlling just one enzyme. It also explains why the lack of a single enzyme can cause disease: in phenylketonuria (PKU), one missing enzyme lets phenylalanine build up to harmful levels.

Anabolic and catabolic reactions

C1.1.3

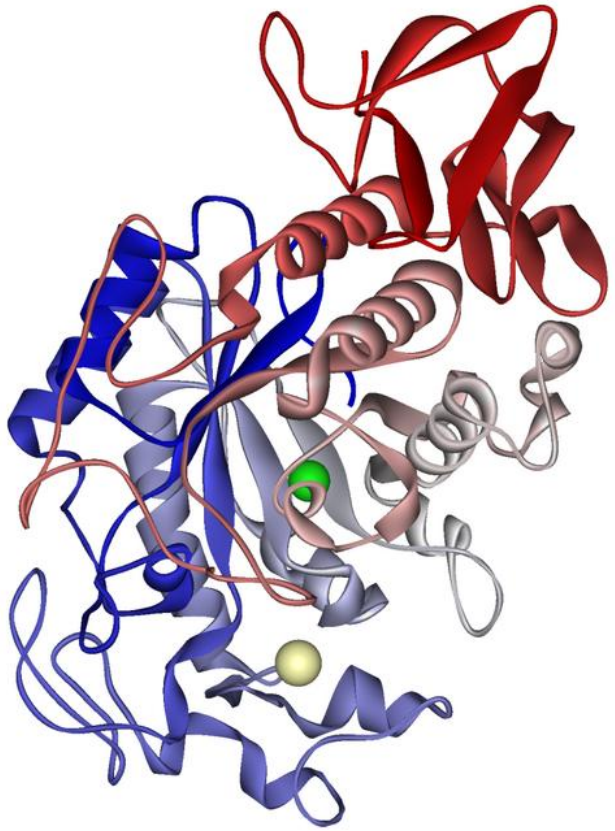
	Anabolic	Catabolic
What happens	Larger molecules are built from smaller ones	Larger molecules are broken into smaller ones
Type of reaction	Usually condensation	Usually hydrolysis or oxidation
Energy	Needs energy, often from ATP	Often releases energy, used to make ATP
Examples	Protein synthesis; photosynthesis; making glycogen	Digestion; cell respiration; breakdown of glycogen

How to remember

Anabolic builds (think of anabolic steroids building muscle). Catabolic breaks down (a catastrophe).

Enzymes: folded proteins with a binding pocket

C1.1.4



Salivary amylase, a globular protein. Public domain.

Enzymes are **globular proteins**: their polypeptide chains fold into a compact, roughly spherical shape.

The **active site** is a small pocket or groove where the substrate binds. Only a few of the enzyme's amino acids form the active site itself; the rest of the protein holds them in exactly the right positions.

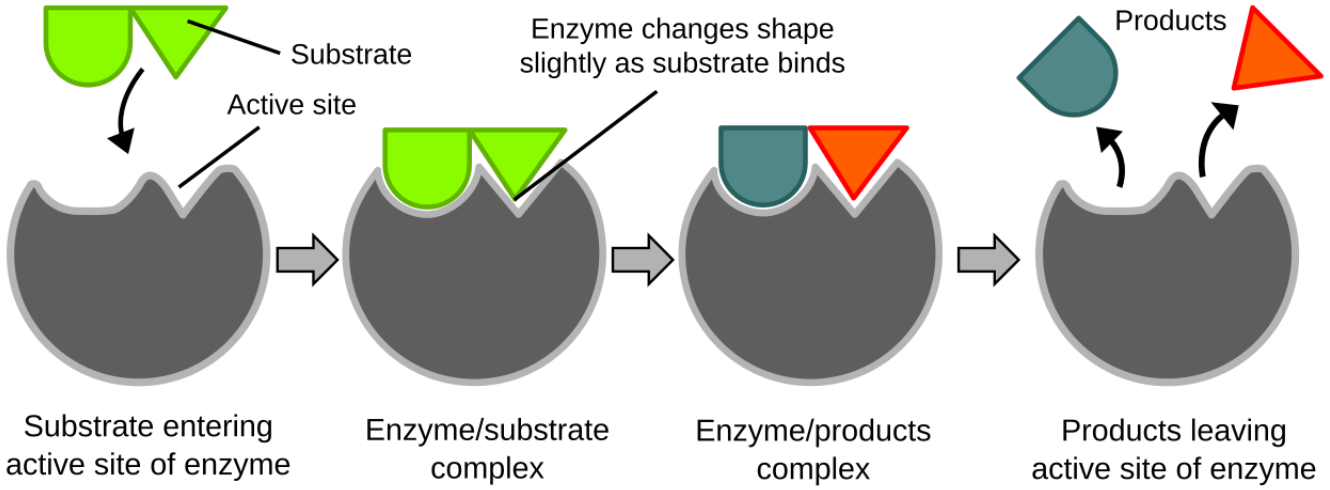
The shape and the chemical properties (charges, polarity) of the active site are **complementary** to the substrate.

Common mistake

The active site and the substrate are not the same shape. Their shapes and chemical properties are complementary, so they fit together.

Induced-fit binding

C1.1.5



Induced fit. TimVickers, vectorized by Fvasconcellos, public domain.

When a substrate enters the active site, **both the substrate and the enzyme change shape** slightly so that they fit more closely. This is the **induced-fit model**.

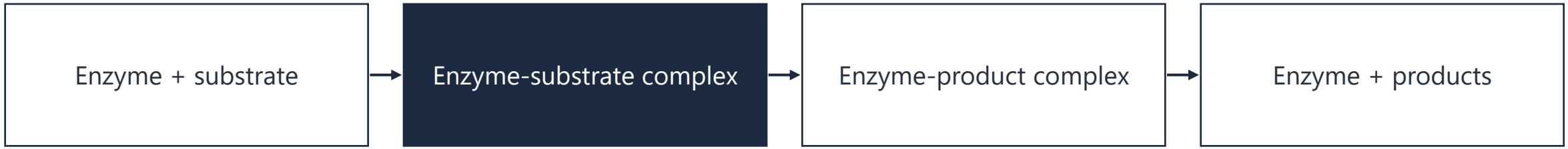
The change in shape puts strain on bonds in the substrate, making them easier to break or form, so the reaction happens more easily.

Compare

An older idea, the lock-and-key model, pictured a rigid active site. Induced fit explains better why enzymes are flexible and why binding speeds the reaction.

The steps of an enzyme-catalysed reaction

C1.1.5



1. The substrate collides with the active site and binds, forming an **enzyme-substrate complex**.
2. Induced fit strains bonds in the substrate, and the reaction takes place while it is bound.
3. The products are released. They no longer fit the active site so well, so they leave.
4. The enzyme is **unchanged** and can bind another substrate molecule. One enzyme can catalyse thousands of reactions every second.

Exam tip

Questions often ask you to outline these stages. Use the terms active site, enzyme-substrate complex and induced fit.

Molecular motion and collisions

C1.1.6

For a reaction to happen, a substrate molecule and an active site must **collide**. Molecules in solution are in constant random motion, so collisions happen all the time, but a collision only leads to binding if the substrate arrives in the **right orientation**.

Situation	What moves
Both dissolved	Both substrate and enzyme move; small substrates move fastest
Enzyme fixed in a membrane	Only the substrate moves to the enzyme
Large, insoluble substrate (such as starch grains)	Only the enzyme moves to the substrate

Key idea

Anything that makes collisions more frequent, such as more substrate or a higher temperature, speeds up the reaction.

Using fixed enzymes in industry

C1.1.6

Enzymes can be **immobilized**: attached to a surface or trapped in beads, so they stay in one place while the substrate flows past.

For example, **lactase** fixed in beads is used to make lactose-free milk. Milk flows through a column of beads; lactose is hydrolysed into glucose and galactose, and the milk leaves with no enzyme in it.

Immobilized enzymes can be reused many times, do not contaminate the product, and are often more stable.

Think about it

In a cell, many enzymes are fixed in membranes, such as those of mitochondria and chloroplasts. Suggest one advantage of keeping the enzymes of a pathway fixed close together. (Hint: think about the product of one step being the substrate of the next.)

Specificity and denaturation

C1.1.7

Term	How it links to the active site
Specificity	Only a substrate with a complementary shape and chemical properties can bind, so each enzyme catalyses one reaction (or a few similar ones)
Denaturation	High temperature or extreme pH breaks the bonds that hold the enzyme's folded shape, so the active site changes shape and the substrate no longer fits

Explain the links

The amino acid sequence decides how the enzyme folds; the fold creates the active site; the active site gives specificity. Change the shape of the active site and both specificity and activity are lost. Denaturation is usually permanent.

Check-in: how enzymes work

C1.1.1–7

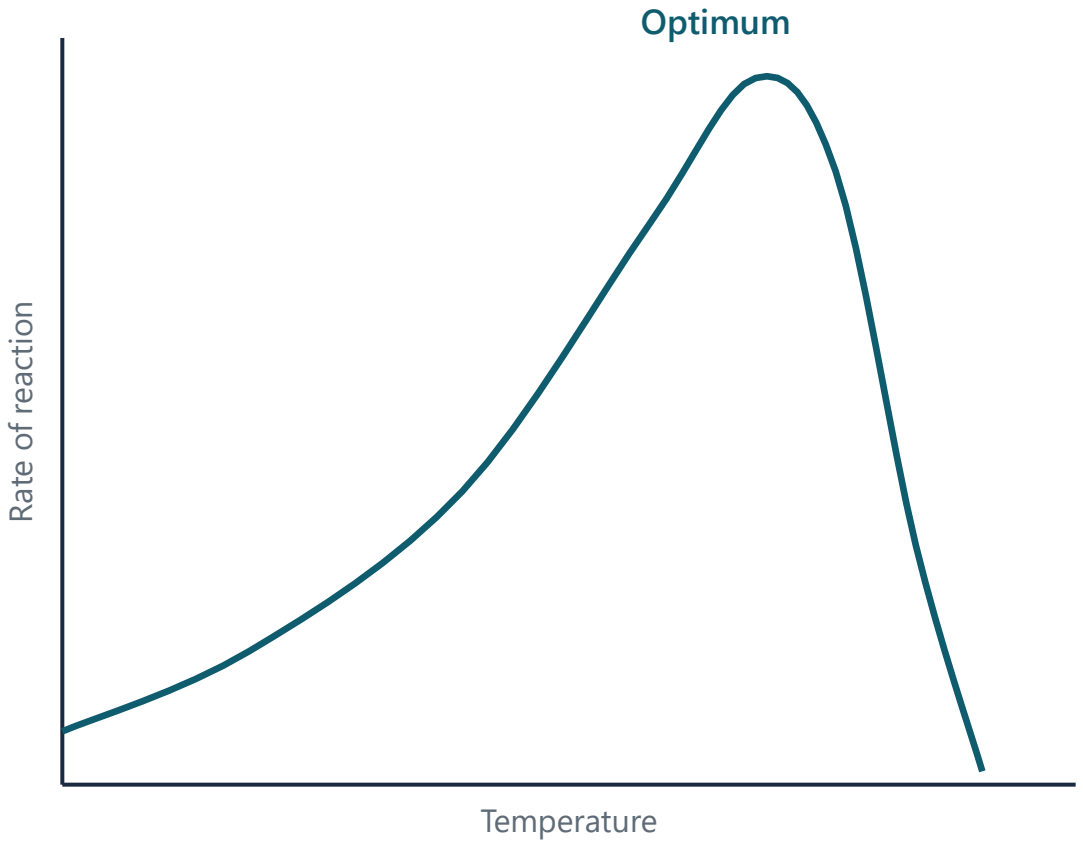
1. Explain why cells need enzymes.
2. Classify as anabolic or catabolic: digestion, protein synthesis, respiration, photosynthesis.
3. Outline the induced-fit model of enzyme action.
4. Explain why an enzyme is specific to its substrate.
5. Explain why immobilized enzymes are useful in industry.

How to use this

Answer without your notes, then compare with a partner.

Effect of temperature

C1.1.8



General shape only, without numerical values.

Rising temperature: molecules move faster, so collisions between substrate and active sites are more frequent and more energetic. The rate rises.

Above the optimum: vibrations break bonds holding the enzyme in shape, the active site starts to change, and the rate falls quickly.

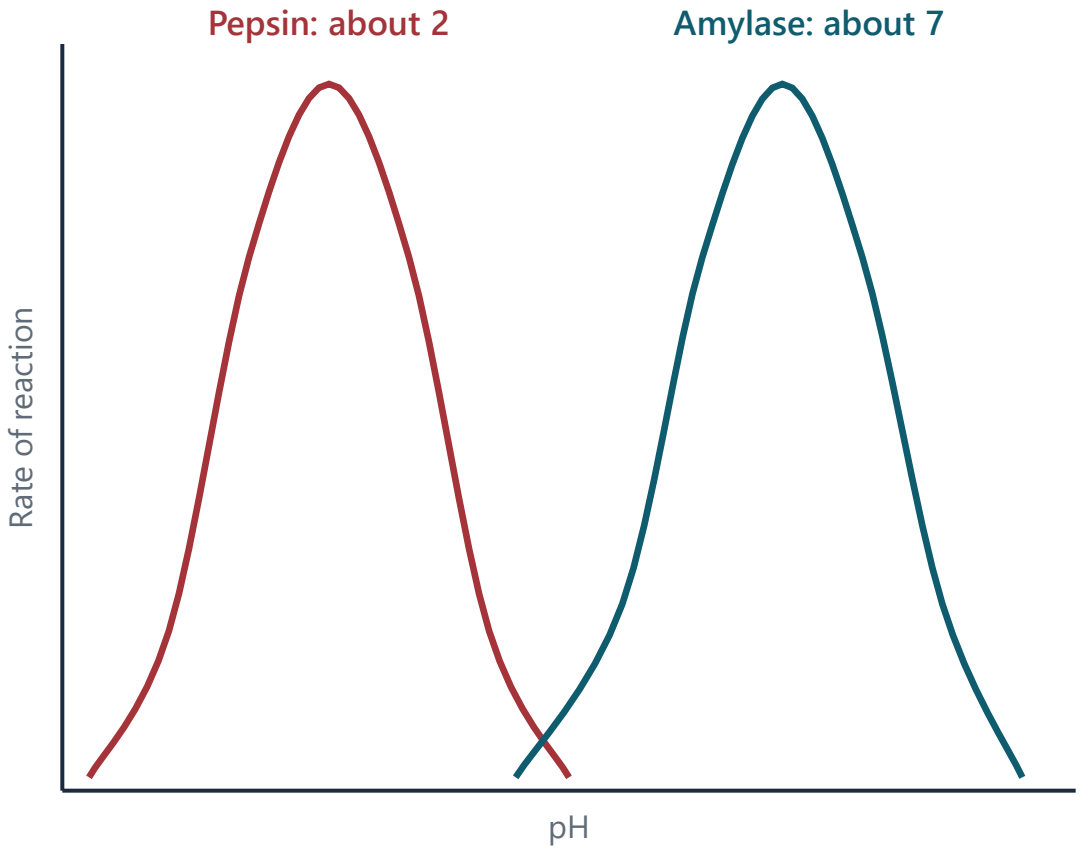
Well above the optimum: the enzyme is fully denatured and the reaction stops.

Big question

A fever of 42 °C is dangerous because vital enzymes begin to denature.

Effect of pH

C1.1.8



Each enzyme works fastest at its **optimum pH**, which usually matches where it works.

On either side of the optimum, H^+ ions are added to or removed from charged R-groups. The attractions holding the active site in shape change, the substrate fits less well, and the rate falls.

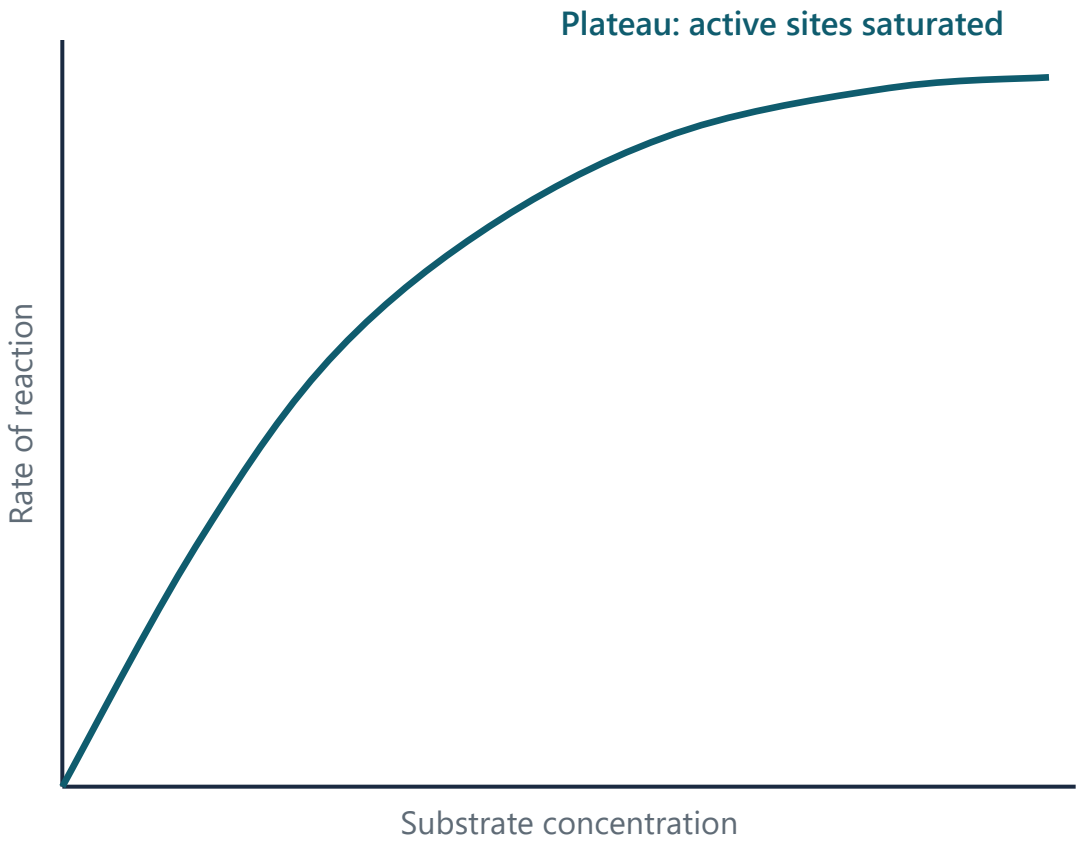
At extreme pH the enzyme is denatured.

Try it

Explain why pepsin works in the stomach but stops working in the small intestine.

Effect of substrate concentration

C1.1.8



Low concentration: few substrate molecules, so collisions with active sites are rare.

Rising concentration: more frequent collisions, so the rate rises.

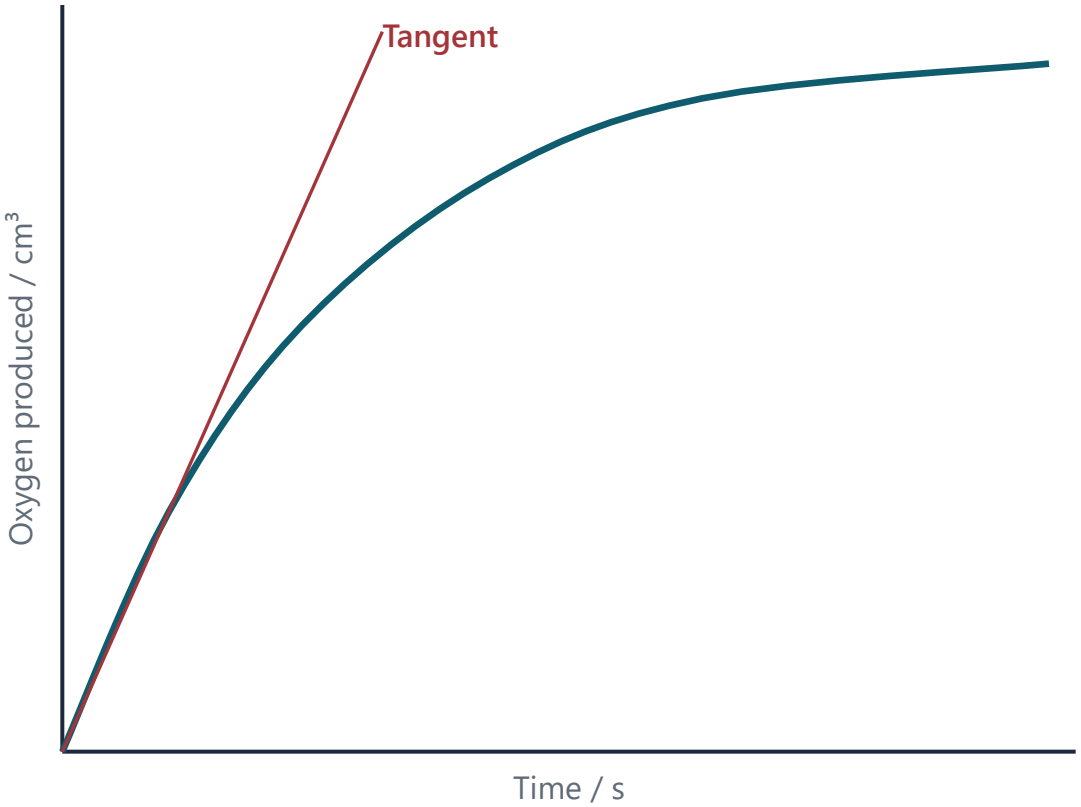
High concentration: almost every active site is occupied all the time. The rate levels off at a maximum, and adding more substrate makes no difference.

Common mistake

There is no optimum here and the enzyme is not denatured: it is working as fast as it can.

Measuring the rate of a reaction

C1.1.9



General shape, for practice.

The rate is the amount of product made (or substrate used) per unit time.

The curve is steepest at the start, when there is most substrate, so experiments compare the **initial rate**: the gradient of a tangent to the curve at time 0.

Worked example

Catalase produced 6 cm^3 of oxygen in the first 20 s, where the graph is almost straight. Initial rate = 6 / 20 = 0.3 cm^3 s^{-1}.

Designing an enzyme experiment

C1.1.8–9

Part of the method	Example: effect of temperature on catalase
Independent variable	Temperature: at least five values, such as 10, 20, 30, 40 and 50 °C, set with a water bath
Dependent variable	Volume of oxygen collected in the first 30 s, by downward displacement of water or a gas syringe
Controlled variables	Volume and concentration of hydrogen peroxide; amount of catalase (such as a 1 cm potato disc); pH (use a buffer); time
Repeats	Three trials at each temperature, then calculate the mean

Safety

Wear eye protection: hydrogen peroxide is an irritant. Do not heat it directly.

Working with data: catalase at three temperatures

C1.1.9

Volume of oxygen released from hydrogen peroxide by the same amount of catalase. These data are invented for practice; the pattern is realistic.

Time / s	0	10	20	30	40	50
20 °C	0	1.2	2.3	3.3	4.2	5.0
35 °C	0	2.6	4.9	6.8	8.3	9.4
60 °C	0	0.6	0.9	1.1	1.2	1.2

1. Calculate the initial rate at each temperature, using the first 10 s.
2. Compare the rates at 20 and 35 °C, and explain the difference.
3. Explain the results at 60 °C.
4. Suggest why the volume levels off at 60 °C after about 40 s.

Units

Volumes of oxygen are in cm^3. Give rates in $cm^3\ s^{-1}$.

Check-in: factors affecting enzymes

C1.1.8–9

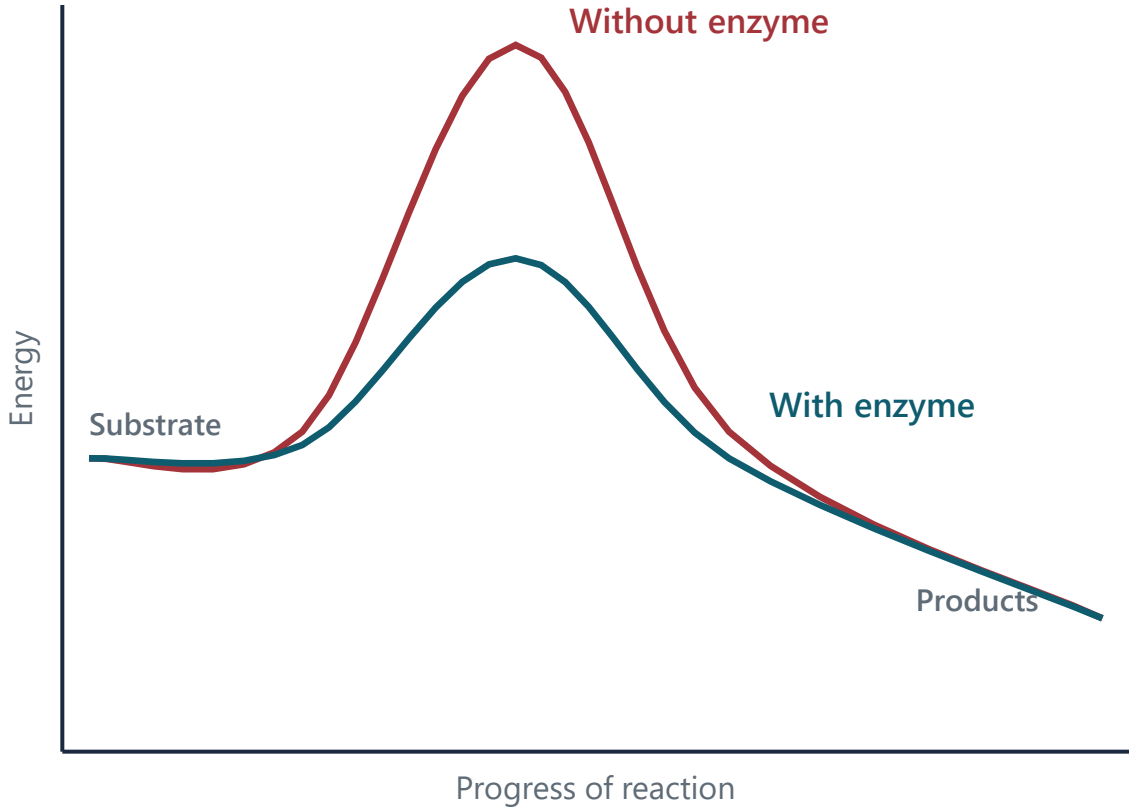
1. Sketch the effect of temperature on the rate of an enzyme-catalysed reaction, and label the optimum.
2. Explain why the rate falls rapidly above the optimum temperature.
3. Explain why the rate levels off at high substrate concentrations.
4. Explain how a change in pH affects an enzyme.
5. State why initial rates are used to compare conditions.

How to use this

Write full sentences, then mark with a partner.

Enzymes lower activation energy

C1.1.10



Activation energy is the energy needed to break or weaken bonds in the substrate so that a reaction can start.

An enzyme **lowers** the activation energy, so at body temperature many more collisions have enough energy to react.

It does **not** change the energy of the substrate or the products, or the overall energy released.

Common mistake

Enzymes do not add energy to a reaction. They lower the energy barrier.

How lowering activation energy speeds up a reaction

C1.1.10

In any group of molecules, a few have a lot of energy and most have an average amount. Only collisions with enough energy to pass the **activation energy** lead to a reaction.

When an enzyme lowers the activation energy, a much larger fraction of collisions have enough energy, so the reaction goes much faster at the **same** temperature.

Induced fit is one way this happens: binding strains bonds in the substrate, so less energy is needed to break them.

Link the ideas

Raising temperature speeds a reaction by giving molecules more energy. An enzyme speeds it up by lowering the energy they need. Cells use enzymes because raising temperature would also denature their proteins.

Common mistakes to avoid

Not this	But this
Enzymes are killed by high temperature.	Enzymes are not alive; they are denatured.
The active site is the same shape as the substrate.	It is complementary in shape and chemical properties.
At high substrate concentration the enzyme is denatured.	The active sites are saturated; the enzyme is working at full speed.
Enzymes give energy to the reaction.	Enzymes lower the activation energy needed.
Low temperature denatures enzymes.	Low temperature slows molecules down; the enzyme is not denatured and works again when warmed.

Check that you can...

1. **Explain** why cells need enzymes as catalysts
2. **Outline** metabolism, and **distinguish** anabolic and catabolic reactions with examples
3. **Describe** the active site, and **explain** induced-fit binding
4. **Explain** the role of molecular motion and collisions, and the use of immobilized enzymes
5. **Explain** specificity and denaturation in terms of the active site
6. **Sketch** and **explain** how temperature, pH and the amount of substrate change the rate
7. **Calculate** initial rates from data, with units
8. **Explain** how enzymes lower activation energy

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Classify each as anabolic or catabolic: digestion, protein synthesis, respiration.
2. Explain why the rate of an enzyme-catalysed reaction falls at very high temperatures.
3. Sketch a graph showing how substrate concentration affects the rate, and label where the active sites are saturated.

Image credits

Glucosidase enzyme.png – Thomas Shafee – CC BY 4.0

Salivary alpha-amylase 1SMD.png – Own work. – Public domain

Induced fit diagram.svg – Created by TimVickers, vectorized by Fvasconcellos – Public domain

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